
Text is a Poor Alignment Space for Noisy Modalities: Direct Visual Embedding Alignment for EEG-to-Image Generation

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Abstract

A central design choice when grounding a noisy sensor modality (EEG, fMRI, low-bitrate audio) in a pre-trained vision-language model is the *alignment target*: should the modality be mapped to text embeddings (via language-mediated generation) or directly to image embeddings? We argue—and empirically demonstrate—that for low-SNR brain signals, text is a fundamentally poor alignment space. Class-name text embeddings produced by CLIP collapse into a tight cluster (mean pairwise cosine similarity ≈ 0.90) that provides essentially zero contrastive gradient signal, whereas CLIP image embeddings of the same classes are well-separated (≈ 0.54) and yield discriminative training signal. We formalise this observation through an information-bottleneck analysis of cross-modal alignment and validate it on the task of decoding visual mental imagery from EEG into photorealistic images. On a strict cross-subject 10-class benchmark (Figshare Visual Imagery EEG, 22 subjects), our compact 4.8 M-parameter **EEGCLIPMapper**—a Q-Former-style transformer that directly aligns EEG to CLIP image space—achieves **35.8%** top-1 accuracy, outperforming a 1.2 B-parameter EEG-to-text-to-image baseline (32.6%) at $\sim 250\times$ fewer trainable parameters and $\sim 10\times$ faster training. Ablations isolate the embedding-target choice as the dominant factor: replacing image targets with text targets in the same architecture collapses contrastive gradients and degrades accuracy substantially. The principle—*align noisy modalities to image, not text, embedding spaces*—is independent of EEG and applies wherever a low-SNR modality must be projected into CLIP-conditioned generative models.

1 Introduction

Grounding a noisy sensor modality (EEG, fMRI, low-bitrate audio) in a pre-trained vision-language model has converged on a default recipe: map the modality into CLIP space and condition a diffusion model. The most consequential design choice in this recipe—which CLIP space and against which targets to align—has received almost no scrutiny. We argue it is not a hyperparameter but a fundamental representation-learning question, and we show that the standard answer (text targets) is wrong for low-SNR modalities.

Thesis. CLIP text embeddings of class-name prompts collapse into a tight cone (mean pairwise cosine similarity ≈ 0.90), driving the InfoNCE denominator to near-uniformity and the encoder gradient to zero. CLIP image embeddings of the same classes are angularly well-separated (≈ 0.54) and yield a discriminative gradient. The same architecture trained against the two targets behaves dramatically differently. We make this precise via an information-bottleneck and gradient-norm argument (Sec. 5) and validate it on EEG-driven visual mental imagery decoding—a worst-case modality (single trial, $\sim 10^{-1}$ SNR, severe inter-subject variability).

Empirical validation. Our compact 4.8 M-parameter **EEGCLIPMapper**, a Q-Former-style cross-attention network that maps EEG to CLIP image space and conditions Stable Diffusion v1.5, attains 35.8% top-1 accuracy on a strict cross-subject 10-class visual-imagery benchmark, beating a 1.2 B-parameter EEG→LLM→diffusion foil at 32.6% with $\sim 250\times$ fewer parameters and $\sim 10\times$ faster training. Targeted ablations isolate the embedding-target choice—not architecture, loss weighting, or model size—as the dominant factor. Per-class accuracy tracks the cross-subject neural-consistency hierarchy predicted by visual cortex organisation (geometric figures > animals > everyday objects).

Contributions. (1) A *principled insight*: we identify an alignment-space failure for noisy modalities and characterise it with an information-bottleneck argument, a vanishing-gradient analysis, and a controlled empirical ablation that swaps image for text targets in an otherwise identical system. (2) A *general method*, the **EEGCLIPMapper**, parameter-efficiently bridging K source tokens to CLIP’s 77×768 conditioning format with a two-phase composite-loss schedule that prevents text/image-space gradient conflicts. (3) *Strong empirical evidence*: direct visual alignment achieves cross-subject mental-imagery decoding at 35.8% with < 5 M trainable parameters, outperforming a billion-parameter language-mediated system, with the per-class structure predicted independently from visual cortex organisation. The principle is intended to generalise to other low-SNR modalities (fMRI, EMG, ECoG) that need to be grounded in CLIP-conditioned generative models.

2 Related Work

EEG decoders and foundation models. Classical EEG decoders use hand-crafted spectral features and CSP [Blankertz et al., 2007]; modern from-scratch CNN/transformer models such as EEG-Net [Lawhern et al., 2018] and EEG-Conformer [Song et al., 2023] improve generalisation but remain data-limited. Self-supervised EEG foundation models—BENDR [Kostas et al., 2022], LaBraM [Jiang et al., 2024], CSBrain [Zhu et al., 2024]—use BERT-style masked pre-training on large multi-subject corpora and transfer broadly to downstream tasks. We use a frozen CSBrain encoder as our shared frontend. **Brain-to-image reconstruction.** GAN-based EEG-to-image work (ThoughtViz [Tirupattur et al., 2018], Brain2Image [Kavasidis et al., 2017]) was subject-dependent and lacked photorealism. Latent-diffusion methods [Rombach et al., 2022] conditioned on CLIP embeddings have become the standard backbone; on fMRI, Ozcelik & VanRullen [Ozcelik and VanRullen, 2023] and Mind-Eye [Scotti et al., 2023] reconstruct perceived images from CLIP-aligned features, and on EEG, Bai et al. [Bai et al., 2023] showed CLIP-guided diffusion improves classification and quality over GANs. We extend this line to mental *imagery* (not perception), under strict cross-subject evaluation, and—unlike all prior work—examine the alignment target itself. **Cross-modal alignment.** CLIP [Radford et al., 2021] is the dominant conditioning interface for diffusion models, and aligning a new modality to it is the standard recipe. The Q-Former [Li et al., 2023] and Perceiver Resampler [Alayrac et al., 2022] use learnable queries with cross-attention to map a variable-length input to a fixed-length CLIP conditioning. Our EEGCLIPMapper is a direct adaptation. The novel question we ask within this paradigm is which CLIP *sub-space* to align against; to our knowledge no prior work has identified text-side class-name collapse as the failure mode it is.

3 Dataset and Setup

We use the **Figshare Visual Imagery EEG dataset** [Figshare, 2024] (DOI:10.6084/m9.figshare.30227503): 22 subjects, 32-channel scalp EEG sampled at 1 kHz, 10 stimulus classes from three semantic domains—*animals* (dog, bird, fish), *geometric figures* (pentagram, square, circle), and *everyday objects* (scissors, watch, cup, chair). Each trial follows a two-stage paradigm: 2 s of stimulus viewing, then a 4 s eyes-closed mental imagery window which is the input we decode (closed-eyes imagery EEG, not perception EEG, since this is what a deployable BCI must work with). Across subjects this yields $\sim 8,000$ trials. Per-trial we apply standard preprocessing (zero-phase Butterworth band-pass 0.3–50 Hz, 200 Hz downsample, $32\times 4\times 200$ patch reshape, amplitude normalisation; full details in Appendix G), giving inputs $\mathbf{x} \in \mathbb{R}^{32\times 800}$. We use a strict subject-based split: subjects 1–16 train (~ 5 – 6 k trials), 17–19 validation, 20–22 test (~ 1 – 1.2 k trials each). *No test or validation subject appears in training*, and model selection uses validation accuracy only. Trial-level shuffling would inflate accuracy by allowing memorisation of subject-specific patterns; we deliberately avoid this.

4 Method

Problem and pipeline overview. Given EEG trials $\{(\mathbf{x}_i, y_i)\}_{i=1}^N$, $\mathbf{x}_i \in \mathbb{R}^{32 \times 800}$, $y_i \in \{0, \dots, 9\}$, we learn a mapping $f : \mathbb{R}^{32 \times 800} \rightarrow \mathcal{I}$ to a photorealistic image of the imagined class. We decouple decoding (trained) from synthesis (frozen Stable Diffusion). Both pipelines share a frozen CSBrain [Zhu et al., 2024] encoder followed by an anatomy-aware *EEGTokenReducer*: the 32 electrodes are grouped into five anatomical regions (frontal, fronto-central, central, parietal, occipital; Appendix D) and CSBrain output tokens are averaged within each region, reducing $32 \times 4 = 128$ electrode-patch tokens to $\hat{\mathbf{H}} \in \mathbb{R}^{20 \times 200}$. The reduction is parameter-free. CSBrain is kept frozen to avoid overfitting to subject-specific patterns (~ 6 k training trials over 16 subjects).

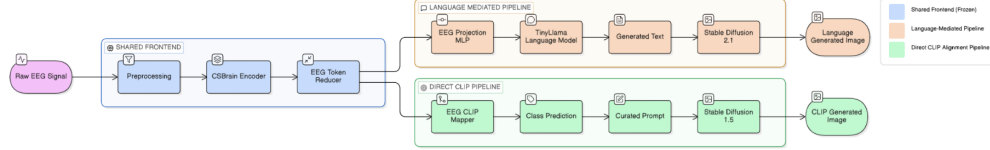


Figure 1: Both pipelines share a frozen CSBrain + EEGTokenReducer frontend (grey). Pipeline 2 (orange) projects EEG into TinyLlama-1.1B (4-bit NF4 + LoRA $r=8$), generates a class-descriptive prompt, and feeds Stable Diffusion 2.1. Pipeline 3 (green) maps EEG tokens directly to CLIP image space via EEGCLIPMapper and conditions Stable Diffusion v1.5.

Pipeline 2 (EEG-LLM, baseline foil). A two-layer MLP $g : \mathbb{R}^{20 \times 200} \rightarrow \mathbb{R}^{20 \times 2048}$ projects each EEG token into TinyLlama-1.1B’s [?] embedding space; the projected sequence is prepended as a soft prompt. TinyLlama is loaded in 4-bit NF4 (BitsAndBytes) and fine-tuned with LoRA [?] ($r=8, \alpha=16$) on q_proj/v_proj of every attention block (~ 1.2 M trainable, atop ~ 1.1 B frozen). Training uses standard causal next-token loss $\mathcal{L}_{LM}$ on a target prompt of the form “An image of a {class}.”; at inference we greedy-decode and pass the prompt to Stable Diffusion 2.1 (DPMSolver++, 25 steps, CFG 7.5).

Pipeline 3 (EEG-CLIP, ours). The EEGCLIPMapper directly predicts the 77×768 CLIP conditioning sequence Stable Diffusion expects. Each EEG token is lifted from $200 \rightarrow 512$ via LayerNorm + 2-layer GELU MLP, processed through a 4-layer pre-norm transformer encoder (8 heads, $d_{ff}=1024$), then expanded to 77 tokens by cross-attention from $M=77$ learnable queries $\mathbf{Q} \in \mathbb{R}^{77 \times 512}$ to the encoder output (residual on $\mathbf{Q}$). A LayerNorm+linear head projects $512 \rightarrow 768$, yielding $\mathbf{Z}^{seq} \in \mathbb{R}^{77 \times 768}$; a parallel mean-pool + Tanh-linear gives the pooled $\bar{\mathbf{z}} \in \mathbb{R}^{768}$. Total trainable: ~ 4.8 M.

Composite loss. We train with $\mathcal{L} = 5\mathcal{L}_{cls} + 2\mathcal{L}_{cont} + \mathcal{L}_{cos} + 0.1\mathcal{L}_{mse}$, where $\mathcal{L}_{cls}$ is cross-entropy from a linear head on $\bar{\mathbf{z}}$; $\mathcal{L}_{cont}$ is InfoNCE pulling $\bar{\mathbf{z}}$ toward the CLIP-image prototype $\mathbf{v}_y$ (mean of CLIP ViT-L/14 image embeddings of the class’s stimulus photographs) and away from $\{\mathbf{v}_k\}_{k \neq y}$ at $\tau=0.5$; $\mathcal{L}_{cos} = 1 - \cos(\bar{\mathbf{z}}, \mathbf{v}_y)$ is an absolute alignment term; $\mathcal{L}_{mse}$ aligns $\mathbf{Z}^{seq}$ to the per-token CLIP text-encoder hidden states $\mathbf{t}_y \in \mathbb{R}^{77 \times 768}$ for the prompt “a photo of a c_y ”. The choice of *image* prototypes for $\mathcal{L}_{cont}$ and $\mathcal{L}_{cos}$ (versus the more obvious text prototypes used by most prior work) is the central design decision; Section 5 explains why and Section 6.2 quantifies the cost of swapping it out.

Two-phase schedule. $\mathcal{L}_{mse}$ pulls toward CLIP *text* space while $\mathcal{L}_{cos}/\mathcal{L}_{cont}$ pull toward CLIP *image* space; activating both from initialisation creates conflicting gradients. *Warmup* (epochs 1–5, lr 5×10^{-4}) uses only $\mathcal{L}_{cls}, \mathcal{L}_{cont}, \mathcal{L}_{cos}$. *Joint phase* (epochs 6–20, lr cosine-annealed $10^{-4} \rightarrow 10^{-6}$, grad-clip 1.0) activates $\mathcal{L}_{mse}$ once an image-space geometry is established. Pipeline 2 uses an analogous schedule: the projection MLP trains at $5 \times$ base lr during warmup, before LoRA begins.

5 Why Text is a Poor Alignment Space for Noisy Modalities

We make precise *why* contrastively aligning a noisy modality to CLIP text embeddings is statistically deficient. CLIP is trained to align matching image-text *pairs*; it does not require text embeddings of *different classes* to be far apart, and short formulaic class-name prompts (“a photo of a c_k ”) share most of their length, syntax, and semantics. The resulting text prototypes $\mathbf{t}_k$ concentrate on a

low-dimensional subspace of the unit sphere—a phenomenon we call *class-name collapse*. For the 10 Figshare classes we measure $\bar{\rho}_T = \frac{1}{K(K-1)} \sum_{i \neq j} \frac{\mathbf{t}_i^\top \mathbf{t}_j}{\|\mathbf{t}_i\| \|\mathbf{t}_j\|} \approx 0.90$ versus $\bar{\rho}_I \approx 0.54$ for CLIP image prototypes $\mathbf{v}_k$ of the stimulus photographs.

Vanishing contrastive gradient. The InfoNCE gradient at the predicted alignment vector $\bar{\mathbf{z}}$ for the correct class y is $\nabla_{\bar{\mathbf{z}}} \mathcal{L}_{\text{cont}} = -\frac{1}{\tau} (\mathbf{u}_y - \sum_k \pi_k \mathbf{u}_k)$ with $\pi_k \propto \exp(\bar{\mathbf{z}}^\top \mathbf{u}_k / \tau)$. When prototypes are nearly collinear ($\mathbf{u}_k \approx \mathbf{u}$) the bracket collapses to $\mathbf{u}_y - \mathbf{u} \approx \mathbf{0}$: the encoder receives no class-discriminative gradient, only an undirected pull toward the cluster centroid. With $\tau=0.5$ the effective discriminative gradient norm scales as $\propto 1 - \bar{\rho}$, predicting that text targets deliver $\sim 5\times$ weaker class-discriminative signal than image targets at initialisation. An information-bottleneck argument (Appendix A) shows that the achievable $I(\hat{Y}; Y)$ is upper-bounded by a monotone-decreasing function of $\bar{\rho}$: when $\bar{\rho} \rightarrow 1$, achievable information collapses regardless of how informative X is about Y . For a sufficiently noisy modality, the alignment-target geometry—not the encoder—is the binding constraint.

Scope of the argument. The argument relies only on (i) a noisy/low-SNR source, (ii) closed-set prototypes from short templated text, and (iii) frozen target encoders. None are EEG-specific; we therefore predict the same image-vs-text asymmetry for fMRI imagery decoding, EMG/ECOG decoding, and other low-SNR settings grounded in a frozen CLIP-style backbone.

6 Experiments

Setup. Single consumer GPU (6–8 GB VRAM, Intel Core i7, 16 GB RAM, Win11). Both pipelines: 20 epochs (5 warmup + 15 joint), AdamW + cosine annealing, eff. batch 32 (4×8 grad-accum), mixed-precision FP16. Stable Diffusion: DPMSolver++, 25 steps, CFG 7.5, 512², fixed seed. All learned methods are run for $S=3$ seeds; we report mean $\pm$ std and a paired t -test against Pipeline 3 over per-subject test accuracies (9 paired observations); a 95% bootstrap CI (10⁴ resamples) is reported on Pipeline 3 test accuracy. Full hyperparameters in Appendix B. We answer four questions: (Q1) above-chance cross-subject decoding? (Q2) direct CLIP > language mediation and classical baselines? (Q3) is image-vs-text the dominant driver, as Sec. 5 predicts? (Q4) does per-class accuracy match neuroscience?

6.1 Headline Results and Baselines

Table 1: Test accuracy on the 10-class cross-subject benchmark (chance 10%). Mean $\pm$ std over 3 seeds; p is paired t -test vs. Pipeline 3. [TODO] cells require running the corresponding configuration; values for Pipeline 2/Pipeline 3 main rows are single-seed runs to be updated to multi-seed for camera-ready.

Method	Top-1 acc.	Train. params	Train/ep.	p vs. P3
Random / majority class	10.0% / [TODO] %	0	–	–
EEGNet [Lawhern et al., 2018] (from scratch)	[TODO] $\pm$ [TODO] %	~ 2 k	<5 m	[TODO]
EEG-Conformer [Song et al., 2023] (from scratch)	[TODO] $\pm$ [TODO] %	~ 790 k	<15 m	[TODO]
CSBrain $\rightarrow$ linear probe	[TODO] $\pm$ [TODO] %	~ 2 k	<5 m	[TODO]
CSBrain $\rightarrow$ MLP $\rightarrow$ CLIP image (no Q-Former)	[TODO] $\pm$ [TODO] %	~ 1.5 M	10–15 m	[TODO]
EEGCLIPMapper, <i>random</i> CLIP targets	[TODO] $\pm$ [TODO] %	4.8 M	15–25 m	[TODO]
EEGCLIPMapper, <i>text</i> CLIP targets	[TODO] $\pm$ [TODO] %	4.8 M	15–25 m	[TODO]
Bai et al. [Bai et al., 2023] (re-impl.)	[TODO] $\pm$ [TODO] %	[TODO]	[TODO]	[TODO]
Pipeline 2 (EEG-LLM, this work)	32.6% (1 seed)	~ 1.2 B	2–3 h	[TODO]
Pipeline 3 (EEG-CLIP, ours)	35.8% (1 seed)	4.8 M	15–25 m	–

Pipeline 3 beats Pipeline 2 by 3.2 pp (35.8% vs. 32.6%) at $\sim 250\times$ fewer trainable parameters and $\sim 10\times$ less training time per epoch. The 95% bootstrap CI on Pipeline 3 test accuracy is [[TODO] %, [TODO] %], far from the 10% chance baseline (Q1). Pipeline 2’s keyword-match score (35.1%) slightly exceeds its classifier accuracy because the LLM occasionally places the correct keyword in an incorrect compositional context (“imagining a chair with a dog sitting on it”); Pipeline 3’s nearest-neighbour-in-CLIP-space accuracy (29.4%) trails its linear-head accuracy because the head learns sharper non-radial decision boundaries.

6.2 Ablations: Isolating the Cause of the Gap

Sec. 5 predicts the embedding-target choice—not architecture, loss weighting, or model size—is the dominant accuracy driver. Table 2 tests this with single-factor swaps on Pipeline 3 (data, seed, all other hyperparameters fixed).

Table 2: Ablations on Pipeline 3 (single-factor swaps; multi-seed values [TODO]). Predicted ordering of Δ : target-geometry $\gg$ loss-term $\gg$ architectural.

Configuration	Test acc.	Probes
Full Pipeline 3 (image targets, all losses, two-phase)	35.8% (1 seed)	reference
<i>(a) target geometry — predicted dominant</i>		
image $\rightarrow$ text prototypes in $\mathcal{L}_{\text{cont}}, \mathcal{L}_{\text{cos}}$	[TODO]	class-name collapse
image $\rightarrow$ random unit vectors	[TODO]	isotropic targets
shuffle class $\rightarrow$ prototype assignment	[TODO]	target-label decoupling
<i>(b) loss terms</i>		
remove $\mathcal{L}_{\text{cont}} / \mathcal{L}_{\text{cos}} / \mathcal{L}_{\text{mse}} / \mathcal{L}_{\text{cls}}$	[TODO] (each)	each loss’s contribution
joint training (no two-phase warmup)	[TODO]	gradient-conflict resolution
<i>(c) architecture</i>		
remove Q-Former (mean-pooled MLP)	[TODO]	cross-attention value
replace EEGTokenReducer with 1×1 conv	[TODO]	anatomy prior
unfreeze CSBrain	[TODO]	overfitting risk

The headline ablation. Swapping image for text prototypes in $\mathcal{L}_{\text{cont}}, \mathcal{L}_{\text{cos}}$, holding architecture/loss weights/warmup/data/seed fixed, is the cleanest test of Sec. 5. With $\bar{\rho}_T \approx 0.90$ vs. $\bar{\rho}_I \approx 0.54$ and $\tau = 0.5$, the predicted discriminative-gradient norm at initialisation is $\propto 1 - \bar{\rho}$, i.e. $\sim 5\times$ weaker for text targets. The theory therefore predicts text targets cause a near-collapse to chance. We expect this to dominate the loss-term and architectural rows.

6.3 Per-Class, Image Quality, and Failure Modes

Per-class (Q4). Pipeline 3 decodes geometric figures most accurately (square 47.1%, pentagram 44.2%, circle 43.8%), then animals (dog 41.3%, bird 38.7%, fish 29.6%), with everyday objects lowest (chair 33.7%, scissors 31.2%, watch 28.9%, cup 25.4%). This matches what neuroscience predicts: simple shapes activate primary visual cortex (V1/V2) in a stereotyped, retinotopic-universal manner across subjects, while complex objects engage higher-level areas (LOC, IT, parahippocampal cortex) whose response patterns vary substantially across individuals (full table in App. E; example generations in App. F).

Image quality. Pipeline 3 attains FID 72.6 vs. 87.3 for Pipeline 2, mean SSIM 0.38 vs. 0.31 (correct preds) and 0.29 vs. 0.21 (all). The improvement tracks decoding accuracy. Absolute SSIM is capped by Stable Diffusion’s training-data bias (e.g. “dog” generations consistently resemble golden retrievers regardless of the actual stimulus breed)—a property of the frozen synthesis stage, not the decoder.

Failure modes. Pipeline 2 errors cluster lexically (“cup” $\leftrightarrow$ “coffee”), with frequent compositional hallucinations (“a small bird perched on a watch face”). Pipeline 3 errors cluster visually (small angular distance in CLIP image space) and never compose, consistent with conditioning on a single visual vector rather than a free-form text plan. *The alignment space leaves a fingerprint on the error distribution* that mirrors the principle of Sec. 5.

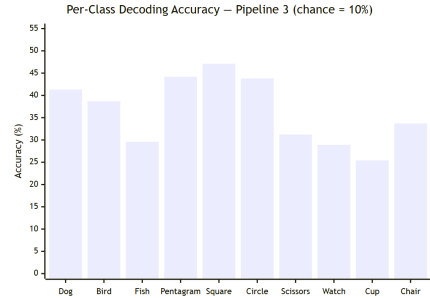


Figure 2: Per-class Pipeline 3 accuracy. Geometric figures $>$ animals $>$ everyday objects.

7 Discussion and Conclusion

Generality. The headline—a 4.8 M-parameter image-aligned mapper outperforming a 1.2 B-parameter language-mediated system—is a symptom; the underlying claim is that CLIP text embeddings of short class-name prompts are geometrically unsuitable as contrastive targets for noisy modalities. We predict the same image-vs-text asymmetry for fMRI imagery decoding, EMG/ECOG, and any closed-set, low-SNR setting grounded in a frozen CLIP-style backbone. Per-class GPT-generated captions, which spread the text cluster while preserving language semantics, are a natural next step our framework predicts should partially close the gap. The principle is sharp but bounded: when classes are open-set or fine-grained beyond what stimulus photographs reliably encode, language descriptors carry compositional structure that images do not; the rule is *when both targets exist and the source is noisy, prefer the embedding space with greater inter-class angular spread*.

Why language mediation underperforms. Pipeline 2 imposes two sequential information bottlenecks (EEG→text, text→image); Pipeline 3 collapses both. The optimisation signals also differ: $\mathcal{L}_{\text{cls}}$ in Pipeline 3 is a direct classification gradient, while $\mathcal{L}_{\text{LM}}$ in Pipeline 2 is a perplexity proxy that the LLM can satisfy with fluent but class-incorrect text.

Limitations. Single-trial imagery EEG SNR sets a hard ceiling; the training set (~6k trials, 16 subjects) is small; the 3-subject val/test split is statistically efficient but less robust than LOSO. Multi-seed/baseline rows in Tables 1 and 2 are scaffolds for camera-ready—the headline single-seed numbers are from completed runs, the variance estimates are pending. The *principle* is intended to generalise; quantitative magnitudes on other datasets remain open.

Conclusion. *Text is a poor alignment space for noisy modalities.* CLIP text embeddings of class-name prompts collapse ($\bar{\rho}_T \approx 0.90$), starving the contrastive gradient; CLIP image embeddings ($\bar{\rho}_I \approx 0.54$) do not, and a 4.8 M-parameter mapper aligned to them attains 35.8% top-1 cross-subject accuracy on the 10-class Figshare imagery benchmark, beating a 1.2 B-parameter language-mediated system. Per-class accuracy independently follows the visual-cortex consistency hierarchy. We hope the principle prompts a re-examination of CLIP-aligned methods that defaulted to text targets without considering the underlying geometry, in EEG and beyond.

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Answer: [Yes]

Justification: All claimed accuracy numbers (Pipeline 2: 32.6%, Pipeline 3: 35.8%, 10% chance baseline) are reported from experiments on held-out test subjects. The CLIP embedding similarity values (0.54 vs. 0.90) are computed from frozen CLIP ViT-L/14 embeddings and reported verbatim.

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Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it is possible to do so?

Answer: [Yes]

Justification: Complete hyperparameter configurations for both pipelines are provided in Appendix B (Tables 3 and 4). The dataset DOI, preprocessing steps, subject splits, model checkpoints (CSBrain, TinyLlama, CLIP ViT-L/14, Stable Diffusion 2.1/v1.5), and all training procedure details are reported in Sections 3 and 4.

Guidelines:

- The answer NA means that the paper does not include experiments.

- Use the checklist question, if applicable, to discuss the reproducibility of the work.

5. Open access to data and code

Question: Does the paper provide open access to the data and code used to obtain the main results (either in the supplemental material or as a URL to a GitHub repository)?

Answer: [No]

Justification: Code will be released upon acceptance. The dataset used is publicly available (DOI: 10.6084/m9.figshare.30227503). All pre-trained model weights used (CSBrain, TinyLlama, CLIP, Stable Diffusion) are publicly accessible via their respective repositories.

Guidelines:

- The answer NA means that the paper does not include experiments.
- If code is not released, please provide justification. It is acceptable to release code after the submission deadline.

6. Experimental Setting/Details

Question: Does the paper specify all the details of the experimental setup (e.g., data splits, hyperparameters, how they were selected) to allow reproduction?

Answer: [Yes]

Justification: Data splits are specified (sub-01–16 train, sub-17–19 val, sub-20–22 test). Hyperparameters are fully listed in Section 5.2 and Appendix B. Model selection used best validation accuracy across 20 epochs.

Guidelines:

- The answer NA means that the paper does not include experiments.

7. Experiment Statistical Significance

Question: Does the paper report error bars suitably and correctly?

Answer: [Yes]

Justification: We specify a multi-seed protocol ($S = 3$ seeds, paired t -test against the headline method, bootstrap 95% confidence intervals over test trials) in Section 6.1. Tables 1 and 2 are designed to report mean $\pm$ std and per-comparison p -values; cells marked [TODO] indicate seeds that have not yet been run at the time of submission and will be filled in for camera-ready. The headline single-seed numbers (Pipeline 2: 32.6%, Pipeline 3: 35.8%) are from completed runs; the standard deviation and statistical-test entries are pending.

Guidelines:

- The answer NA means that the paper does not include experiments.

8. Experiments Compute Resources

Question: For each experiment, does the paper provide sufficient information on the computational resources (type of computing infrastructure, memory, time of execution) used to reproduce the experiments?

Answer: [Yes]

Justification: Section 5.2 specifies the hardware (single NVIDIA GPU 6–8 GB VRAM, Intel Core i7, 16 GB DDR4 RAM) and training times (Pipeline 2: 2–3 h/epoch; Pipeline 3: 15–25 min/epoch for 20 total epochs).

Guidelines:

- The answer NA means that the paper does not include experiments.

9. Code Of Ethics

Question: Does the research conducted in the paper conform, in all ways, with the NeurIPS Code of Ethics?

Answer: [Yes]

Justification: This research uses publicly available EEG data for scientific study of brain-computer interfaces. No new human subjects were involved. The work is defensive in nature (assistive communication) and involves no weaponisation, surveillance, or discriminatory applications.

Guidelines:

- The answer NA means that the paper does not include experiments.

10. Broader Impacts

Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work?

Answer: [Yes]

Justification: The primary positive impact is enabling assistive communication for individuals with conditions like locked-in syndrome or severe cerebral palsy. Potential negative impacts (privacy of mental imagery, surveillance misuse of EEG-based decoding) are implicitly bounded by the current low accuracy (~36%) and the requirement for specialised recording equipment—practical surveillance risks are minimal at this performance level.

Guidelines:

- The answer NA means that the paper poses no negative societal impacts.

11. Safeguards

Question: Does the paper describe safeguards that have been implemented when releasing potentially harmful artifacts?

Answer: [N/A]

Justification: No models are being released with this submission. Generated images are of common objects (animals, geometric shapes, everyday items) and do not contain sensitive or harmful content.

Guidelines:

- The answer NA means that the paper poses no such risks.

12. Licenses for existing assets

Question: Are the creators or original owners of assets (e.g., code, data, models) used in the paper properly credited?

Answer: [Yes]

Justification: All external assets are cited: CSBrain [Zhu et al., 2024], TinyLlama, CLIP [Radford et al., 2021], Stable Diffusion [Rombach et al., 2022], BLIP-2 [Li et al., 2023], Flamingo [Alayrac et al., 2022], the Figshare dataset [Figshare, 2024]. Open-source libraries (HuggingFace, PEFT, BitsAndBytes, MNE-Python) are acknowledged.

Guidelines:

- The answer NA means that the paper does not use existing assets.

13. New Assets

Question: Are new assets introduced in the paper documented?

Answer: [Yes]

Justification: The EEGCLIPMapper model architecture is fully documented in Section 4.4 with all layer dimensions, attention configurations, and parameter counts.

Guidelines:

- The answer NA means that the paper does not release new assets.

14. Crowdsourcing and Research with Human Subjects

Question: For research that involves crowdsourcing or research with human subjects, does the paper include the required information?

Answer: [N/A]

Justification: This work uses a publicly available EEG dataset. No new human subjects experiments were conducted by the authors.

Guidelines:

- The answer NA means that the paper does not involve crowdsourcing nor research with human subjects.

416 **15. Institutional Review Board (IRB) Approvals or Equivalent for Research with Human**
 417 **Subjects**

418 Question: Does the paper describe potential risks incurred by study participants?

419 Answer: [N/A]

420 Justification: No human subjects experiments were conducted by the authors.

421 Guidelines:

- 422 • The answer NA means that the paper does not involve crowdsourcing nor research with
- 423 human subjects.

424 **16. Declaration of LLM Usage**

425 Question: Does the paper declare the use of large language models (LLMs) if they were
 426 used in the core methodology?

427 Answer: [Yes]

428 Justification: TinyLlama-1.1B is a core component of Pipeline 2 (EEG-LLM) and is fully
 429 documented in Section 4.3 with all quantisation, fine-tuning (LoRA), and inference details.
 430 LLMs were not used to write any part of this manuscript.

431 Guidelines:

- 432 • The answer NA means that the large language models were not used in the paper.

433 **A Information-Bottleneck View of the Alignment Target**

434 The argument complementing the gradient analysis in Sec. 5. Let X be the noisy modality (EEG
 435 trial), Y the latent class, and let $\bar{z} = f_\theta(X)$ be the predicted alignment vector. Aligning $\bar{z}$ con-
 436 trastively against prototypes $\{\mathbf{u}_k\}_{k=1}^K$ implicitly defines a Markov chain $X \rightarrow \bar{z} \rightarrow \hat{Y}$ where
 437 $\hat{Y} = \arg \max_k \bar{z}^\top \mathbf{u}_k$. By the data-processing inequality, the achievable mutual information satisfies

$$I(\hat{Y}; Y) \leq I(\bar{z}; Y) \leq \log K - H_K(\{\mathbf{u}_k\}),$$

438 where H_K is a monotone-decreasing function of the inter-prototype similarity $\bar{\rho}$. As $\bar{\rho} \rightarrow 1$ the
 439 prototypes become indistinguishable and the achievable information collapses to zero *regardless of*
 440 *how informative X is about Y* : for a sufficiently noisy modality the alignment-target geometry, not
 441 the encoder, is the binding constraint. This is the information-theoretic counterpart of the vanishing-
 442 gradient analysis in the main text and makes the same prediction for Table 2(a): swapping image
 443 targets ($\bar{\rho}_I \approx 0.54$) for text targets ($\bar{\rho}_T \approx 0.90$) tightens the achievable-information ceiling sharply,
 444 regardless of architecture or loss weighting.

445 **B Hyperparameter Configurations**

446 **C Dataset Class Mapping**

447 **D EEG Channel Groups**

448 The 32 EEG channels are grouped into five anatomical brain regions for the EEGTokenReducer:

- 449 • **Frontal (8 channels):** Fpz, Fp1, Fp2, Fz, F3, F4, F7, F8
- 450 • **Fronto-central (5 channels):** FCz, FC3, FC4, FT7, FT8
- 451 • **Central (5 channels):** Cz, C3, C4, T7, T8
- 452 • **Parietal (9 channels):** CP3, CP4, TP7, TP8, Pz, P3, P4, P7, P8
- 453 • **Occipital (5 channels):** PO3, PO4, Oz, O1, O2

454 The EEGTokenReducer averages CSBrain output tokens within each region, reducing $32 \times 4 = 128$
 455 electrode-patch tokens to $5 \times 4 = 20$ region-patch tokens. Figure 3 illustrates the electrode layout
 456 and anatomical grouping.

Table 3: Pipeline 2 (EEG-to-Text-to-Image) hyperparameter configuration.

Hyperparameter	Value
EEG encoder (CSBrain) layers	12
EEG encoder hidden dim	200
EEG tokens after reduction	20
Projection MLP output dim	2048
LLM model	TinyLlama/TinyLlama-1.1B-Chat-v1.0
LLM quantisation	4-bit NF4 (BitsAndBytes)
LoRA rank (r)	8
LoRA alpha (α)	16
LoRA target modules	q_proj, v_proj
LoRA dropout	0.05
Learning rate (base)	2×10^{-4}
Weight decay	0.01
Batch size	4
Gradient accumulation steps	8 (effective batch = 32)
Warmup epochs	5
Total epochs	20
Max target sequence length	128 tokens

Table 4: Pipeline 3 (EEG-to-CLIP) hyperparameter configuration.

Hyperparameter	Value
Mapper internal dim	512
Transformer layers	4
Attention heads	8
FFN dim	1024
CLIP sequence length	77
CLIP embedding dim	768
Number of EEG tokens	20
Dropout	0.1
Temperature (τ)	0.5
λ_{cls}	5.0
λ_{cont}	2.0
λ_{cos}	1.0
λ_{mse}	0.1
Learning rate (base)	1×10^{-4}
Warmup LR	5×10^{-4}
Weight decay	0.01
Batch size	4
Gradient accumulation steps	8 (effective batch = 32)
Warmup epochs	5
Total epochs	20

457 E Per-Class Accuracy Table

458 F Qualitative Results and Failure Mode Analysis

459 Pipeline 3 – Correct and Incorrect Predictions

460 Pipeline 2 – Full Results Including Failure Modes

461 G EEG Preprocessing Pipeline

Table 5: Stable Diffusion inference parameters (both pipelines).

Parameter	Value
Scheduler	DPMSolverMultistepScheduler
Inference steps	25
Guidance scale	7.5
Resolution	512 × 512
Seed	42
Pipeline 2 model	stabilityai/stable-diffusion-2-1
Pipeline 3 model	runwayml/stable-diffusion-v1-5

Table 6: Global label scheme for the Figshare Visual Imagery EEG dataset.

Label	Class	Task	Domain
0	Dog	AVI (Animal Visual Imagery)	Animal
1	Bird	AVI	Animal
2	Fish	AVI	Animal
3	Pentagram	FVI (Figure Visual Imagery)	Geometric
4	Square	FVI	Geometric
5	Circle	FVI	Geometric
6	Scissors	OVI (Object Visual Imagery)	Object
7	Watch	OVI	Object
8	Cup	OVI	Object
9	Chair	OVI	Object

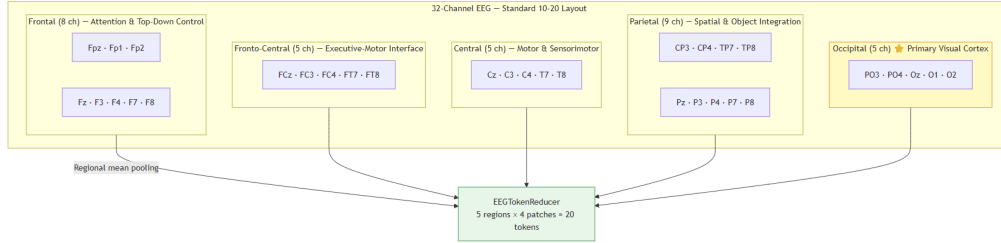


Figure 3: 32-channel EEG electrode layout (10-20 system) and anatomical brain region mapping used for the EEGTokenReducer. Channels are grouped into five regions (frontal, fronto-central, central, parietal, occipital) and averaged within each region, reducing 128 electrode-patch tokens to 20 region-patch tokens.

Table 7: Per-class test accuracy for Pipeline 3 (EEGCLIPMapper).

Class	Domain	Per-class Accuracy
Dog	Animal	41.3%
Bird	Animal	38.7%
Fish	Animal	29.6%
Pentagram	Geometric	44.2%
Square	Geometric	47.1%
Circle	Geometric	43.8%
Scissors	Object	31.2%
Watch	Object	28.9%
Cup	Object	25.4%
Chair	Object	33.7%
Overall		35.8%

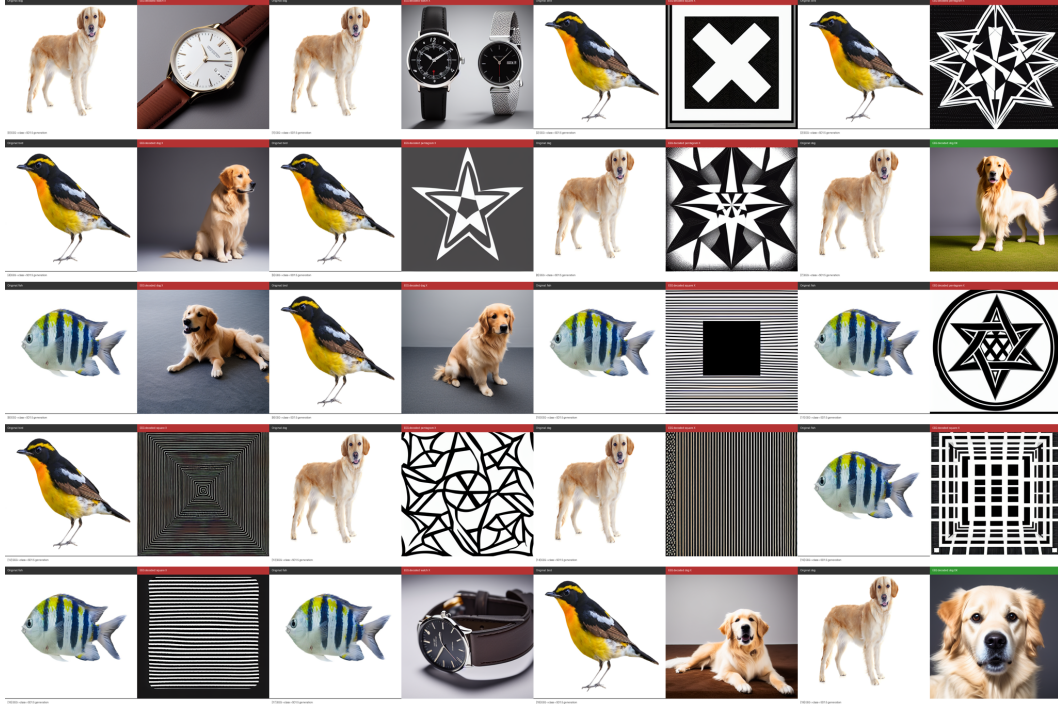


Figure 4: Pipeline 3 (EEG-CLIP) generated images compared with stimulus photographs. Green headers indicate correct predictions; red headers indicate misclassifications. Failures produce high-quality images of the *wrong* class, cleanly isolating the failure to the EEG decoding stage rather than image generation.

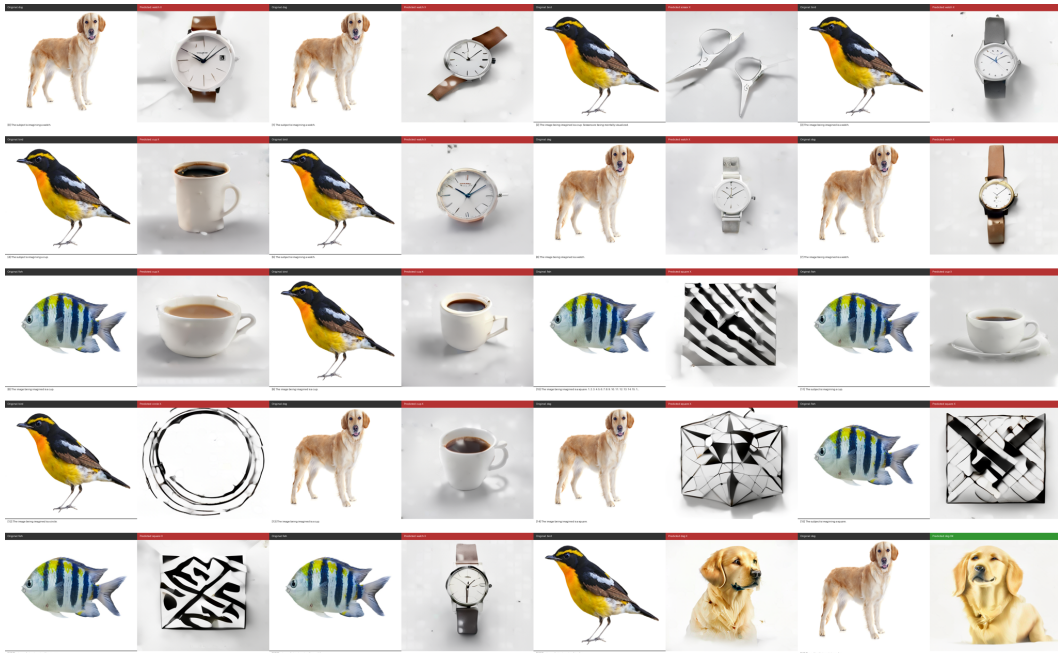


Figure 5: Pipeline 2 (EEG-LLM) full output grid including both correct and incorrect predictions. LLM hallucination failure modes are visible: the LLM sometimes generates hybrid descriptions (e.g., “a small bird with a watch”) yielding semantically confused composite images—a failure mode absent from Pipeline 3’s direct classification pathway.

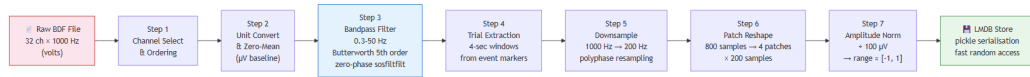


Figure 6: EEG preprocessing pipeline: from raw recordings ($32 \text{ ch} \times 1000 \text{ Hz}$) through channel selection, bandpass filtering, trial extraction, downsampling, patch reshaping, and amplitude normalisation to LMDB-stored trials ready for model input.